IYYANAR S

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SUMMARY

Biomedical Engineering student with hands-on experience in Medical Imaging, Biomedical Signal Processing, AI, and Medical Device Development. Skilled in deep learning, image processing, physiological signal analysis, and embedded healthcare systems. Driven to engineer intelligent, innovative, and clinically relevant healthcare technologies.

EDUCATION

BE-BME | PSG College of Technology, Coimbatore | CGPA (8.01) — Expected in May 2027
CLASS 12 | Sri Lakshmi Vidhyalaya Matric Hr Sec School, Villupuram | Score (91.1%) — March 2023

SKILLS

-**Programming Languages:** Python, C, SQL
-**Soft Skills:** Teamwork, Communication, Organisation, Leadership, Event Management
-**Biomedical Testing:** PCB Designing, Component Testing, Quality Inspection, Troubleshooting, Documentation
-**Embedded & Hardware:** Arduino, Raspberry Pi, ESP 32, Altium Designer, Multisim, PSpice, Kinovea, COMSOL
-**Tools:** GitHub, Adobe Photoshop & Illustrator, Excel, PyTorch

INTERNSHIP

Biomedical Engineer| Medex Technologies Pvt. Ltd., Coimbatore — *18 May 2026 –20 June 2026*
Worked hands-on with anaesthesia machine assembly, PCB design in Altium Designer, component selection, soldering and desoldering, and hardware integration. Also gained practical exposure to PCB testing, quality checks, and the overall medical device development process**.**

Biomedical Engineer Trainee | Popular Nursing Home, Bangalore — *8 Dec 2025 – 13 Dec 2025*
Assisted with equipment rounds and learned to troubleshoot biomedical devices. Gained understanding of biomedical equipment in patient care. Observed laparoscopic surgery procedures

PROJECTS

-**Development of a Non-Contact Heart Rate and Respiration Rate for an Emergency Clinical Setting**: Designed a contactless vital sign monitoring system using mmWave radar for real-time Heart Rate (HR) and Respiration Rate (RR) estimation through chest motion sensing. Applied signal processing techniques, including phase extraction, filtering, FFT analysis, and HR/RR band separation for accurate physiological monitoring. Developed a real-time healthcare monitoring solution to improve patient comfort, reduce infection risk, and support emergency care applications.

-**Thyroid Nodule Detection & Segmentation:** Developed an AI-based thyroid ultrasound image segmentation system using U-Net with ResNet34 backbone on the TN3K dataset for accurate thyroid nodule detection and mask prediction. Implemented preprocessing, custom PyTorch data pipelines, Grad-CAM visualisation, and optimised training using Adam optimiser, BCE with LogitsLoss, and Dice Coefficient evaluation. Achieved reliable segmentation performance for medical image analysis using Python, PyTorch, OpenCV, and deep learning techniques.

-**Automated Colorectal Polyp Detection & Segmentation – A Deep Learning System for Early Colorectal Cancer Prevention:** Developed an automated colorectal polyp detection and segmentation system using a U-Net deep learning architecture on the Kvasir-SEG dataset with Dice + BCE hybrid loss for accurate early-stage colorectal cancer prediction and medical image segmentation.

-**Contactless Patient Monitoring System using PIR and Temperature Sensors:** Developed a contactless patient monitoring system using Arduino UNO, PIR, and LM35 sensors for real-time body temperature and motion detection with automated buzzer and LED alerts for abnormal patient conditions.

POSITION AND RESPONSIBILITY

-**Design Team Head**, Biomedical Engineering Association, PSG College Of Technology
-**Design Team Head**, IEEE Students Chapter 12951, PSG College Of Technology
-**Publicity Team Member**, IETE Students' Chapter at PSG Technology
-**Placement representative,** BE- Biomedical Engineering, Batch (2023-2027), PSG College of Technology

CERTIFICATIONS

-MATLAB onramp (**MATLAB**)
-Image processing on-ramp (**MATLAB**)
-Introduction to Deep Learning (**GUVI**)
-IOT Application Development with the ESP32 (**Udemy**)
-Medical Image Processing (**Coursera**)
-PCB Basic Design Course (**Altium Education**)
-Java and JavaScript (**Simplilearn**)

HACKATHONS & WORKSHOPS

Origin Medical Imaging AI Hackathon 2026
AI-Powered Embedded Systems, Product Prototyping for Industry Applications

EXTRACURRICULAR ACTIVITIES

Photography, Video Editing, Playing Cricket, E-sports, Fitness and Strength Training.